

Diagnostic report: Mg I

Computed levels: 30

Matched to NIST: 16

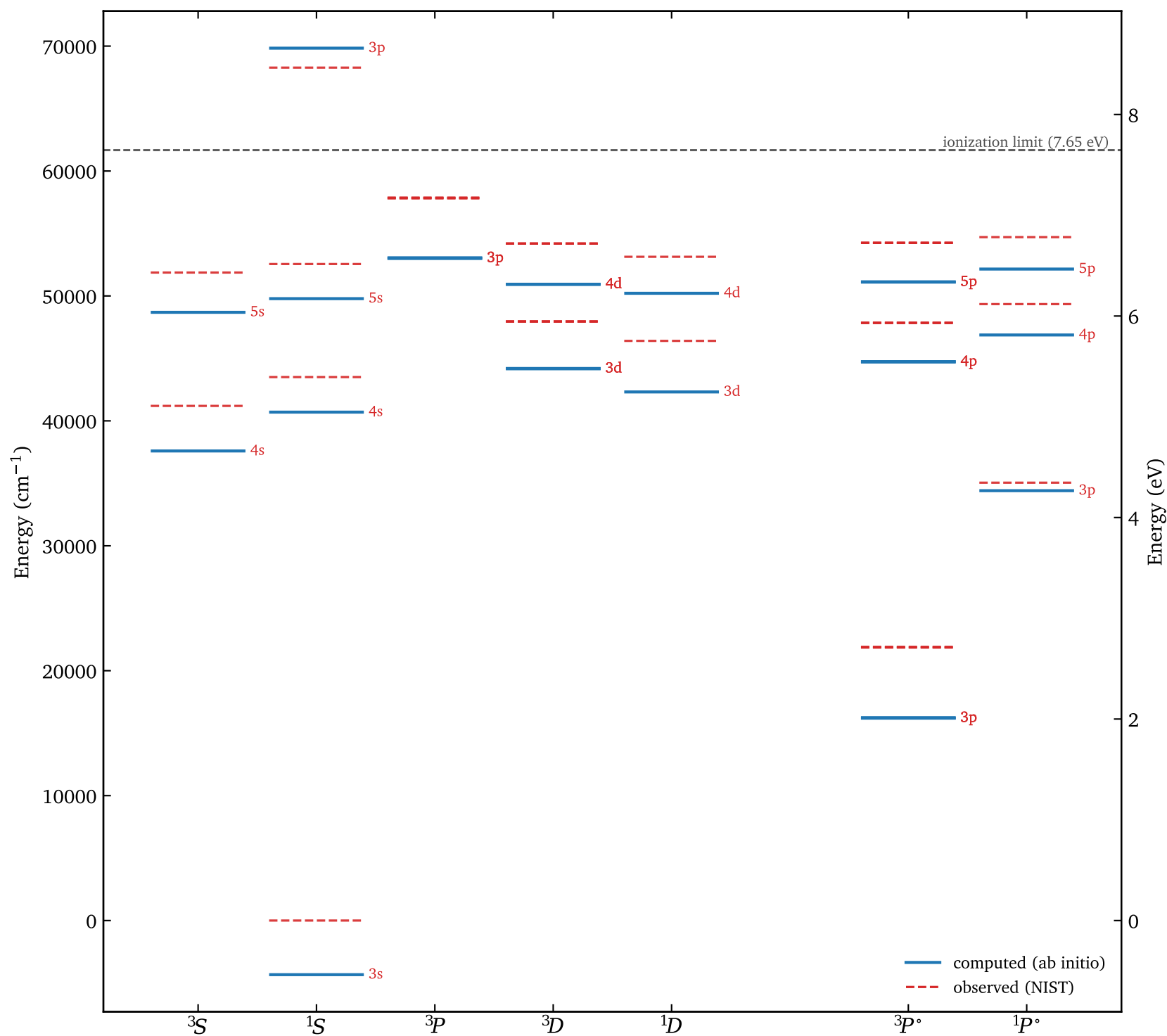
Computed E1 lines: 66

Pre-fit RMS ($E_{\text{calc}} - E_{\text{obs}}$): 29252 cm^{-1} (ab initio, no RCE fit yet)

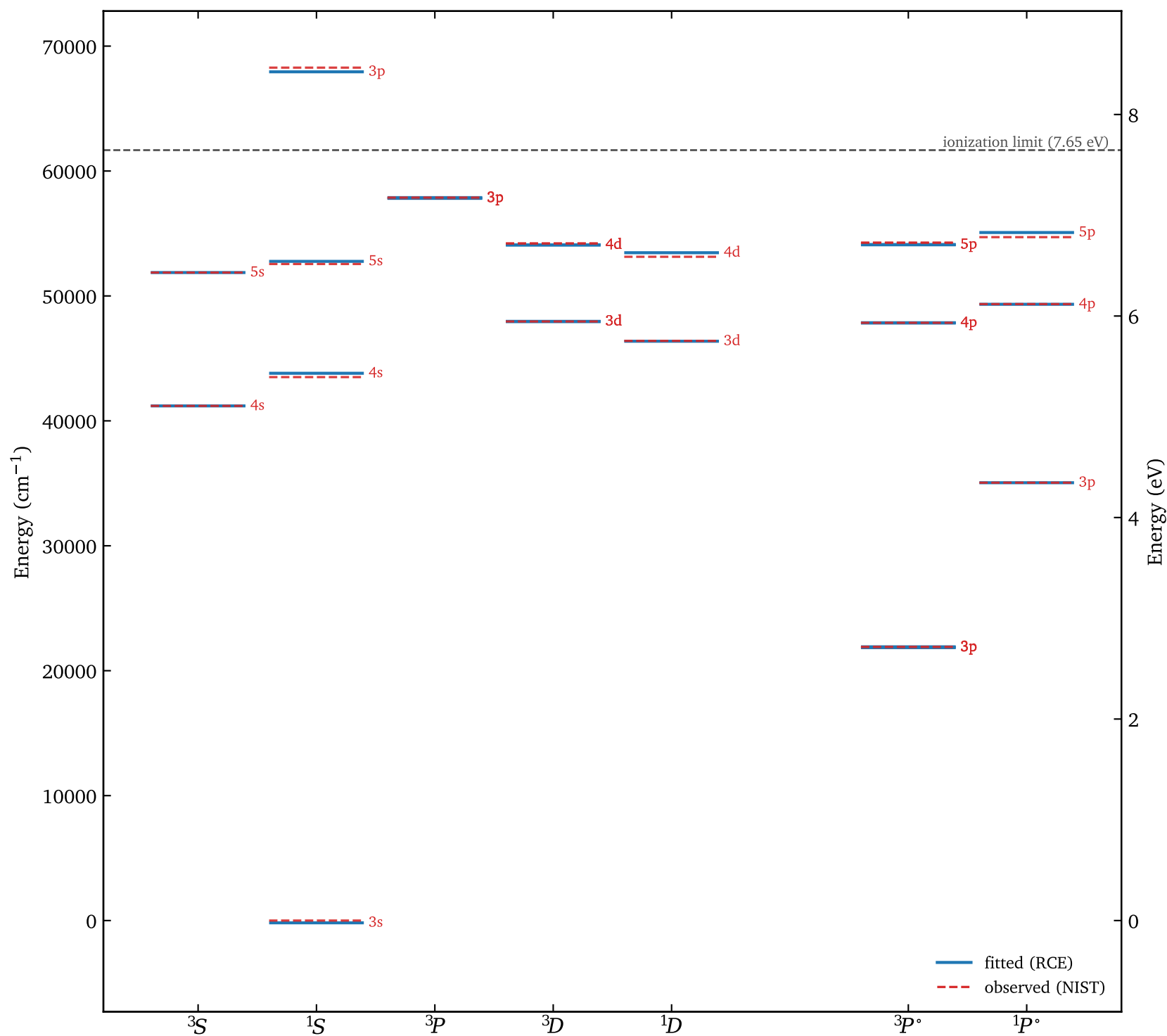
Strongest computed lines:

lambda(A)	log gf	upper term
3566.85	+0.789	(2S) 1P
29958.38	+0.521	(2S) 1P
41200.71	+0.509	(2S) 3P
2716.41	+0.470	(2S) 3P
16124.69	+0.438	(2S) 3P
51567.09	+0.429	(2S) 1P
13993.58	+0.406	(2S) 3P
3578.48	+0.365	(2S) 3P

Mg I: term diagram - ab initio vs observed



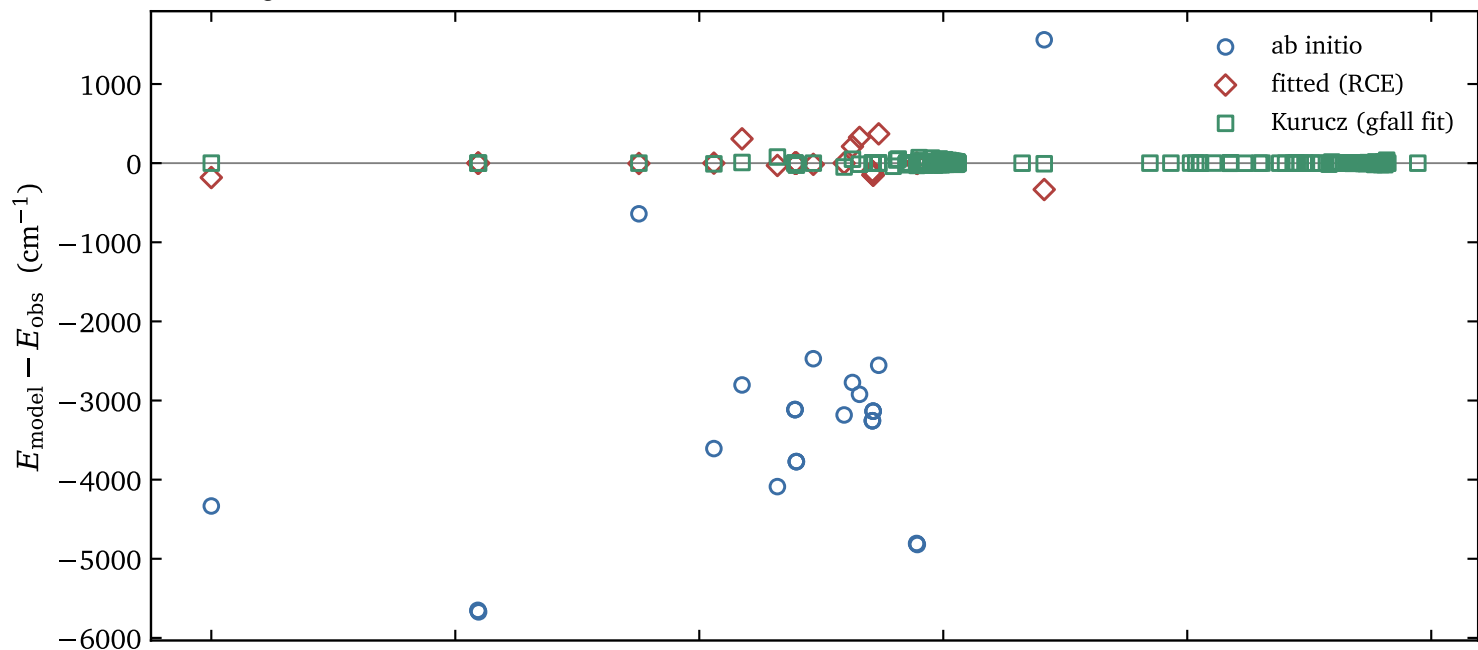
Mg I: term diagram - fitted vs observed



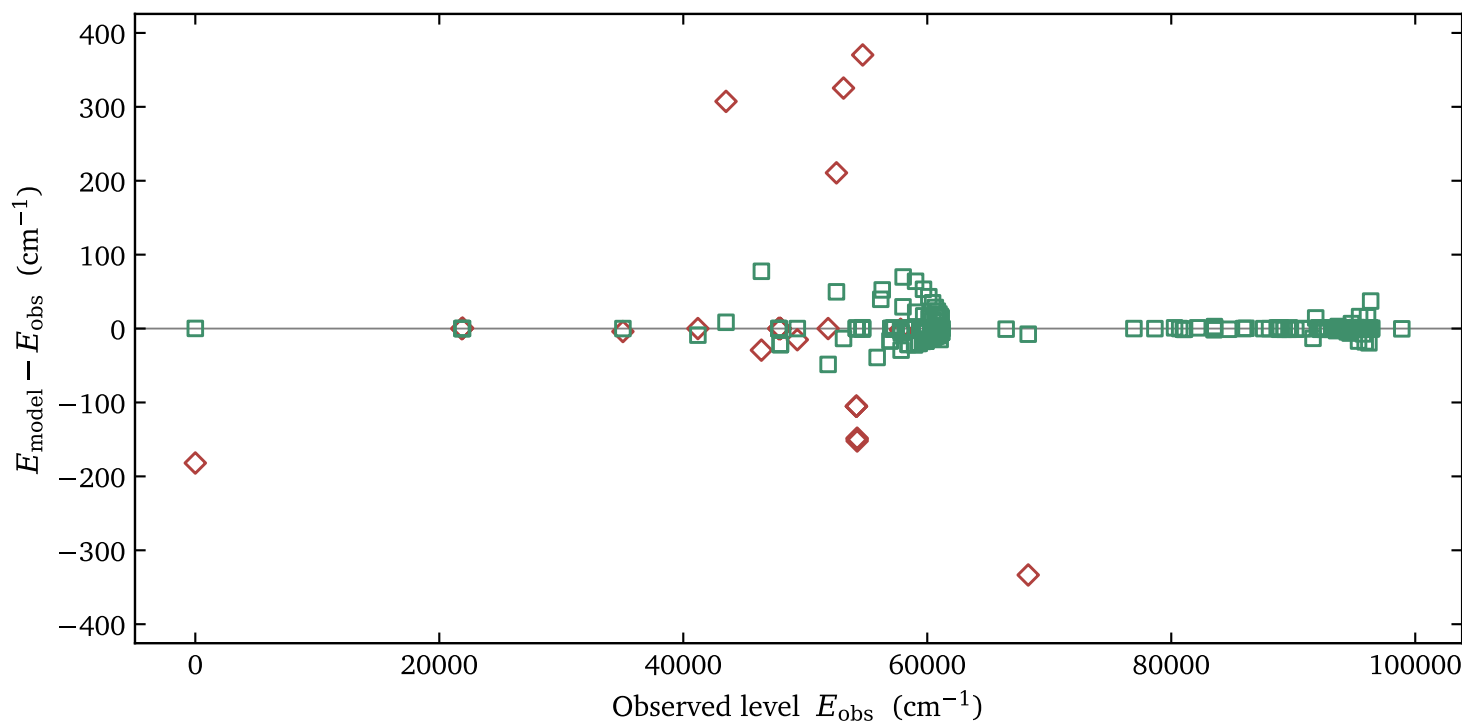
Mg I: levels — ab initio & fitted vs observed (cm^{-1})

config	term	J	E_obs	E_calc	Δ_{abinit}	E_fit	Δ_{fit}
3s2	1S	0	0.0	-4333.0	-4333.0	-181.9	-181.9
3s.3p	$^3P^\circ$	0	21850.4	16199.0	-5651.4	21850.2	-0.2
3s.3p	$^3P^\circ$	1	21870.5	16213.0	-5657.5	21870.7	+0.2
3s.3p	$^3P^\circ$	2	21911.2	16240.0	-5671.2	21911.8	+0.6
3s.3p	$^1P^\circ$	1	35051.3	34411.0	-640.3	35047.3	-4.0
3s.4s	3S	1	41197.4	37590.0	-3607.4	41197.4	+0.0
3s.4s	1S	0	43503.3	40700.0	-2803.3	43810.7	+307.4
3s.3d	1D	2	46403.1	42315.0	-4088.1	46373.9	-29.2
3s.4p	$^3P^\circ$	0	47841.1	44727.0	-3114.1	47841.0	-0.1
3s.4p	$^3P^\circ$	1	47844.4	44730.0	-3114.4	47844.5	+0.1
3s.4p	$^3P^\circ$	2	47851.2	44736.0	-3115.2	47851.4	+0.2
3s.3d	3D	1	47957.1	44185.0	-3772.1	47957.1	+0.0
3s.3d	3D	2	47957.0	44185.0	-3772.0	47957.1	+0.1
3s.3d	3D	3	47957.0	44185.0	-3772.0	47957.1	+0.1
3s.4p	$^1P^\circ$	1	49346.7	46875.0	-2471.7	49331.6	-15.1
3s.5s	3S	1	51872.5	48690.0	-3182.5	51872.5	+0.0
3s.5s	1S	0	52556.2	49784.0	-2772.2	52766.9	+210.7
3s.4d	1D	2	53134.6	50213.0	-2921.6	53460.0	+325.4
3s.4d	3D	1	54192.3	50937.0	-3255.3	54087.2	-105.1
3s.4d	3D	2	54192.3	50937.0	-3255.3	54087.2	-105.1
3s.4d	3D	3	54192.3	50937.0	-3255.3	54087.2	-105.1
3s.5p	$^3P^\circ$	0	54248.8	51114.0	-3134.8	54100.5	-148.3
3s.5p	$^3P^\circ$	1	54250.1	51115.0	-3135.1	54100.5	-149.6
3s.5p	$^3P^\circ$	2	54252.7	51118.0	-3134.7	54100.6	-152.1
3s.5p	$^1P^\circ$	1	54706.5	52152.0	-2554.5	55076.7	+370.2
3p2	3P	0	57812.8	53006.0	-4806.8	57811.8	-1.0
3p2	3P	1	57833.4	53022.0	-4811.4	57831.5	-1.9
3p2	3P	2	57873.9	53054.0	-4819.9	57870.3	-3.6
3p2	1S	0	68275.0	69834.0	+1559.0	67941.6	-333.4

Mg I: level residuals (ab initio RMS = 3699, fitted RMS = 147, Kurucz RMS = 12 cm⁻¹)



zoom to fitted residuals



Mg I: gf comparison vs NIST
ab initio: strong RMS=0.11 (all 0.19)
fitted (energy+gf): strong RMS=0.09 (all 0.22)
Kurucz (gfall): strong RMS=0.82 (all 0.53) [dex; strong = $\log gf \geq -1$]

